



# MAARULA CLASSES

**TARGET- NIMCET / CUET.PG By: Amit Katiyar (MCA-JNU)**

**SET THEORY**  
**NIMCET PYQ**



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1. In a class, 50 student play cricket, 20 play hockey, 10 play both the games, then the no. of students playing at least one game is:

(a)30 (b)60 (c)70 (d)none

**NIMCET 2007**

2. If A and B are two sets, then  $A' - B'$

(a) $B - A$  (b) $A - B$  (c) $\phi$  (d)none

**NIMCET 2007**

3. Suppose  $P_1, P_2, \dots, P_{30}$  are thirty sets each having 5 elements and  $Q_1, Q_2, \dots, Q_n$  are n sets with 3 elements each. Let  $\bigcup_{i=1}^{30} P_i = \bigcup_{j=1}^n Q_j = S$  and each element of S belongs to exactly 10 of the  $P_i$ s and exactly 9 of  $Q_j$ s. then n is equal to

(a)15 (b)3 (c)45 (d) None

**NIMCET 2008**

4. If  $P = \{(4^n - 3n - 1) | n \in N\}$  and  $Q = \{(9n - 9) | n \in N\}$ , then  $P \cup Q$  is equal to

(a)N (b)P (c)Q (d)None

**NIMCET 2009**

5.  $A_1, A_2, A_3$  and  $A_4$  are subsets of a set U containing 75 elements with the following properties:

Each subset contains 28 elements; the intersection of any two of the subsets contains 12 elements, the intersection of any three of the subsets contains 5 elements; the intersection of all four subsets contains 1 element. The number of elements belonging to none of the subsets is

(a)15 (b)17 (c)16 (d)18

**NIMCET 2009**

6. From 50 students taking examination in Mathematics, Physics and Chemistry, 37 passed Mathematics, 24 Physics, and 43 Chemistry. Atmost, 19 passed Mathematics and physics, atmost 29 Mathematics and Chemistry and atmost 20 physics and Chemistry. The largest possible number of that could have passed all three examinations is

(a)10 (b)12 (c)9 (d)None

**NIMCET 2009**

7. How many proper subsets of  $\{1, 2, 3, 4, 5, 6, 7\}$  contain the numbers 1 and 7?

(a)7 (b)31 (c)32 (d)62

**NIMCET 2010**

8. Identify the wrong statement from the following.

**NIMCET 2010**

(a) If A and B are two sets, then  $\overline{A \cup B} = \overline{A} \cap \overline{B}$

(b) If A, B and C are sets, then

$$(A - C) - C = (A - C) - (B - C)$$

(c) If A and B are two sets, then  $\overline{A \cup B} = \overline{A} \cap \overline{B}$

(d) If A, B and C are sets, then  $A \cap B \cap \overline{C} \subseteq A \cap B$

9. Let X be the universal set for sets A and B. if  $n(A) = 200, n(B) = 300$  and  $n(A \cap B) = 100$ , then  $n(A' \cap B')$  is equal to 300. Provided  $n(X)$  is equal to

**NIMCET 2011**

(a) 600 (b)700 (c)800 (d)900

10. In a college of 300 students, every student reads 5 newspapers and every newspaper is read by 60 students. The number of newspaper is

**NIMCET 2011**

(a) atleast 30 (b) atmost 20  
(c) exactly 25 (d) exactly 28

11. In a class of 100 students, 55 students have passes in Mathematics and 67 students have passed in Physics. Then, the number of students who have passed in Physics only, is

**NIMCET 2012**

(a)22 (b)33 (c)10 (d)45

12. Find the number of elements in the union of four sets A, B, C and D having 150, 10, 210 and 240 elements respectively, given that each pair of sets has 15 elements in common. Each triplets of sets has 3 elements in common and  $A \cap B \cap C \cap D = \phi$

**NIMCET 2013**

(a) 616 (b) 512 (c)111 (d) 702

13. If the sets A and B are defined as

**NIMCET 2014**

$$A = \{(x, y) | y = 1/x, 0 \neq x \in R\},$$

$$B = \{(x, y) | y = -x, x \in R\}, \text{ then}$$

(a) $A \cap B = \phi$  (b) $A \cap B = B$

(c) $A \cap B = A$  (d)None of these

14. A professor has 24 text books on computer science and is concerned about their coverage of the topics (P) compilers, (Q) data structure and (R) operating system. The following data gives the number of books that contain material on these topics  
 $n(P) = 8, n(Q) = 13, n(R) = 13, n(P \cap Q) = 3, n(P \cap Q \cap R) = 2$ , where  $n(x)$  is the cardinality of the set x. then, the number of text books that have no material on compiler is

**NIMCET 2015**

(a)4 (b)8 (c)12 (d)16

15. Let  $\overline{P}$  and  $\overline{Q}$  denote the complements of two sets P and Q. then, the set  $(P - Q) \cup (Q - P) \cup (P \cap Q)$  is

**NIMCET 2015**

(a) $P \cup Q$  (b) $\overline{P} \cup \overline{Q}$  (c) $P \cap Q$  (d) $\overline{P} \cap \overline{Q}$





16. If  $X = \{4^n - 3n - 1, n \in \mathbb{N}\}$  and  $Y = \{9n - 9, n \in \mathbb{N}\}$ , then  $X \cup Y$  is equal to **NIMCET 2016**

- (a)Y (b)X (c)N (d)None

17. Which of the following statements is FALSE? **NIMCET 2016**

- (a)  $2 \in A \cup B$  implies that if  $2 \notin A$ , then  $2 \in B$   
(b)  $\{2, 3\} \subseteq A$  implies that  $2 \subseteq A$  and  $\{2, 3\} \subseteq B$   
(c)  $A \cup B \supseteq \{2, 3\}$  implies that  $\{2, 3\} \subseteq A$  and  $\{2, 3\} \subseteq B$   
(d)  $\{2\} \in A$  and  $\{3\} \in A$  implies that  $\{2, 3\} \subseteq A$

18. Two finite sets A and B are having m and n elements. The total numbers subsets of the first set is 56 more than the total total number of subsets of the second set. The value of m and n are **NIMCET 2016**

- (a)7, 6 (b)6, 3 (c)5, 3 (d)8, 7

19. Let  $S = \{1, 2, \dots, n\}$ . The form  $(A, B)$  with  $A \subseteq B$  for subsets A, B of S is **NIMCET 2016**

- (a) $2^n$  (b) $3^n$   
(c) $n!$  (d) $\sum_{k=0}^n \binom{n}{k} \left(\frac{n}{n-k}\right)$

20. The number of elements in the power set  $P(S)$  of the set  $S = \{2, (1, 4)\}$  is **NIMCET 2017**

- (a)2 (b)4 (c)8 (d)10

21. Let A and B two sets containing four and two elements, respectively. The number of subsets of the  $A \times B$ , each having atleast three element is **NIMCET 2017**

- (a)270 (b)239 (c)219 (d)256

22. In a survey where 100 students reported which subjects they like, 32 students in total liked Mathematics, 38 students in total liked Business and 30 students liked Literature. Moreover 7 students liked both Mathematics and Literature, 10 students liked both Mathematics and Business, 8 students liked both Business and Literature, 5 students liked all three subjects. Then, the number of people who liked exactly one subject is **NIMCET 2018**

- (a)60 (b)65 (c)70 (d)78

23. Suppose  $A_1, A_2, \dots, A_{30}$  are thirty set each having 5 elements with no common element across the sets and  $B_1, B_2, \dots, B_n$  are n sets each having 3 elements with no common element across the sets. Let  $\bigcup_{i=1}^{30} A_i = \bigcup_{j=1}^n B_j = S$ , and each element of S belongs to exactly 10 of  $A_i$ 's and exactly 9 of  $B_j$ 's. The value of n is equal to **NIMCET 2019**

- (a)15 (b)30 (c)40 (d)45

24. In a class of 50 students, it was found that 30 students read "Hitavad", 35 students read "Hindustan" and 10 read neither. How many students read both "Hitavad" and "Hindustan" newspaper? **NIMCET 2020**

- (a)25 (b)20 (c)15 (d)30

25. If  $A = \{4^x - 3x - 1 : x \in \mathbb{N}\}$  and  $B = \{9(x - 1) : x \in \mathbb{N}\}$ , where N is the set of natural number, then **NIMCET 2020**

- (a) $A \subset B$  (b) $A \subseteq B$  (c) $A \supset B$  (d) $A \supseteq B$

26. If  $A = \{x, y, z\}$ , then the number of subsets in power set of A is **NIMCET 2020**

- (a)6 (b)8 (c)7 (d)9

27. If A is a subset of B and B is a subsets of C, then cardinality of  $A \cup B \cup C$  is equal to **NIMCET 2020**

- (a) Cardinality of C (b) Cardinality of B  
(c) Cardinality of A (d) None of these

28.  $A = \{1, 2, 3, 4\}$ ,  $B = \{3, 4, 5\}$  then the number of elements in  $(A \cup B) \times (A \cap B) \times (A \Delta B)$  is **NIMCET 2021**

- (a)18 (b)20 (c)24 (d)30

29. Suppose  $A_1, A_2, \dots, A_{30}$  are thirty set each having 5 elements with no common element across the sets and  $B_1, B_2, \dots, B_n$  are n sets each having 3 elements with no common element across the sets. Let  $\bigcup_{i=1}^{30} A_i = \bigcup_{j=1}^n B_j = S$ , and each element of S belongs to exactly 10 of  $A_i$ 's and exactly 9 of  $B_j$ 's. The value of n is equal to **NIMCET 2021**

- (a)15 (b)30 (c)40 (d)45

30. If X and Y are two sets, then  $X \cap Y' \cap (X \cup Y)'$  is **NIMCET 2021**

- (a) $X'$  (b) $Y'$  (c) $\phi$  (d)None

31. There are two sets A and B with  $|A| = m$  and  $|B| = n$ . If  $|P(A)| - |P(B)| = 112$ , then choose the wrong options (where  $|A|$  denotes the cardinality of A, and  $P(A)$  denotes the power set of A) **NIMCET 2022**

- (a) $m + n = 11$  (b) $2n - m = 1$   
(c) $2m - n = 1$  (d) $3n - m = 5$

32. A survey is done among a population of 200 people who likes either tea or coffee. It is found that 60% of the population like tea and 72% of the population like coffee. Let x be the number of people who like both tea and coffee. Let  $m \leq x \leq n$ , then choosen the correct option. **NIMCET 2022**

- (a)  $n - m = 56$  (b)  $n - m = 28$   
(c)  $n - m = 32$  (d)  $n + m = 92$



33. Let A and B be sets,  $A \cap X = B \cap X = \phi$  and  $A \cup X = B \cup X$  for some set X, relation between A and B.

**NIMCET 2023**

- (a)  $A = B$  (b)  $A \cup B = X$   
(c)  $B = X$  (d)  $A = X$

34. Let Z be the set of all integers, and consider the sets  $X = \{(x,y) : x^2 + 2y^2 = 3, x, y \in Z\}$  and  $Y = \{(x,y) : x > y, x, y \in Z\}$ . Then the number of elements in  $X \cap Y$  is (Set Theory)

- (a) 2 (b) 1 (c) 3 (d) 4

**NIMCET-2024**

35. Let C denote the set of all tuples (x,y) which satisfy  $x^2 - 2y = 0$  where x and y are natural numbers. What is the cardinality of C? (Set Theory)

**NIMCET-2024**

- (a) 0 (b) 1 (c) 2 (d) 3

36. Out of a group of 50 students taking examinations in Mathematics, Physics, and Chemistry, 37 students passed Mathematics, 24 passed Physics, and 43 passed Chemistry. Additionally, no more than 19 students passed both Mathematics and Physics, no more than 29 passed both Mathematics and Chemistry, and no more than 20 passed both Physics and Chemistry. What is the maximum number of students who could have passed all three examinations? (Set Theory)

**NIMCET-2024**

- (a) 14 (b) 12  
(c) 10 (d) 17

37. Find the cardinality of the set C which is defined as

$$C = \left\{x \mid \sin 4x = \frac{1}{2} \text{ for } x \in (-9\pi, 3\pi)\right\} \text{ (Set Theory)}$$

- (a) 24 (b) 48 (c) 36 (d) 12

**NIMCET-2024**

38. Let  $A = \{5^n - 4n - 1 : n \in \mathbb{N}\}$  and  $B = \{16(n - 1) : n \in \mathbb{N}\}$  be sets. Then

**NIMCET-2025**

- (a) Neither  $A \subset B$  nor  $B \subset A$   
(b)  $A \subset B$   
(c)  $B \subset A$   
(d)  $A \cap B$  is a finite set

39. Suppose that C represents the set of all countries, R represents the set of all countries that have at least one river flowing through it, M represents the set of all countries that have at least one mountain in it, and D represents the set of all countries that have at least one desert in it. It is given that  $(R \cup M \cup D) = C$  Which one of the following gives the set of all countries that have either a mountain or a river, but does not have a desert in it? The notation  $D^c$  represents the complement of the set D with respect to the universal set C.

**NIMCET-2025**

- (a)  $(R \cup M) - (R \cap M \cap D^c)$   
(b)  $(R \cap M) \cap D^c$   
(c)  $(R \cup M) \cap D^c$   
(d)  $(R \cup M) - (R \cap M)$

## ANSWER KEY

| 1.  | 2.  | 3.  | 4.  | 5.  | 6.  | 7.  | 8.  | 9.  | 10. |
|-----|-----|-----|-----|-----|-----|-----|-----|-----|-----|
| b   | a   | c   | c   | c   | d   | b   | b   | b   | c   |
| 11. | 12. | 13. | 14. | 15. | 16. | 17. | 18. | 19. | 20. |
| d   | d   | a   | d   | a   | a   | *   | b   | b   | b   |
| 21. | 22. | 23. | 24. | 25. | 26. | 27. | 28. | 29. | 30. |
| c   | b   | d   | a   | a   | b   | a   | d   | b   | c   |
| 31. | 32. | 33. | 34. | 35. | 36. | 37. | 38. | 39. |     |
| c   | a   | a   | a   | c   | d   | b   | b   | c   |     |

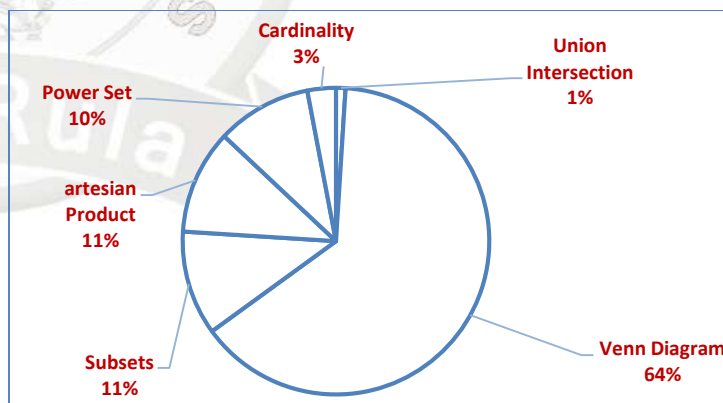
**Note (\*)** Wrong

## Analysis 2007 to 2025

### Questions Asked in NIMCET 2007 to 2025

| Years       | No. of Questions | Years       | No. of Questions |
|-------------|------------------|-------------|------------------|
| NIMCET 2007 | 02               | NIMCET 2016 | 04               |
| NIMCET 2008 | 01               | NIMCET 2017 | 02               |
| NIMCET 2009 | 03               | NIMCET 2018 | 01               |
| NIMCET 2010 | 02               | NIMCET 2019 | 01               |
| NIMCET 2011 | 02               | NIMCET 2020 | 04               |
| NIMCET 2012 | 01               | NIMCET 2021 | 03               |
| NIMCET 2013 | 01               | NIMCET 2022 | 02               |
| NIMCET 2014 | 01               | NIMCET 2023 | 01               |
| NIMCET 2015 | 02               | NIMCET 2024 | 04               |
|             |                  | NIMCET 2025 | 02               |

## SUB TOPIC ANALYSIS



## Sub Topicwise Analysis

| Sub Topic          | Question Asked |
|--------------------|----------------|
| Venn Diagram       | 18             |
| Subsets            | 03             |
| Cartesian Product  | 03             |
| Power Set          | 03             |
| Cardinality        | 01             |
| Union Intersection | 01             |

## SOLUTION

1. (b)  
2. (a)





3. (c)  $n(P_1) + n(P_2) + \dots + n(P_{30}) = 5 \times 30 = 150$   
 $n(Q_1) + n(Q_2) + \dots + n(Q_n) = 3 \times n = 3n$   
 Let set  $S$  has  $k$  element then,  
 $k = \frac{150}{10} \Rightarrow 15$  and  $k = \frac{3n}{9} = \frac{n}{3} \Rightarrow 15 = \frac{n}{3} \Rightarrow n = 45$

4. (c) For,  $n = 1$   
 $P = \{4 - 3 - 1\} = \{0\}, Q = \{9 - 9\} = \{0\}$   
 For,  $n = 2$   
 $P = \{4^2 - 3 \times 2 - 1\} = \{16 - 6 - 1\} = \{9\}$   
 $Q = \{9 \times 2 - 9\} = \{18 - 9\} = \{9\}$   
 For,  $n = 3$   
 $P = \{4^3 - 3 \times 3 - 1\} = \{54\}$   
 $Q = \{9 \times 3 - 9\} = \{18\}$   
 $P = \{0, 9, 54, \dots\} Q = \{0, 9, 18, \dots\}$   
 So, clearly  $P$  is a subset of  $Q$ , so  $P \cup Q = Q$

5. (c)  $n(U) - n(A_1 \cup A_2 \cup A_3 \cup A_4)$   
 $= \sum n(A_i) - \sum n(A_i \cap A_j) + \sum n(A_i \cap A_j \cap A_k) - n(A_1 \cap A_2 \cap A_3 \cap A_4)$   
 Total two pair from  $A_1, A_2, A_3, A_4 = {}^4C_2 = 6$   
 Total triplets from  $A_1, A_2, A_3, A_4 = {}^4C_3 = 4$   
 So,  $n(A_1 \cup A_2 \cup A_3 \cup A_4)$   
 $= (4 \times 28) - (6 \times 12) + (4 \times 5) - 1112$   
 $-72 + 20 - 1 = 59$   
 Element belonging to none of four set  
 $= n(U) - n(A_1 \cup A_2 \cup A_3 \cup A_4) = 75 - 59 = 16$

6. (d)  $n(M \cup P \cup C) = 50$   
 $n(M) = 37$   
 $n(P) = 24$   
 $n(C) = 43$   
 $n(M \cap P) \leq 19$   
 $n(M \cap C) \leq 29$   
 $n(P \cap C) \leq 20$   
 $n(M \cup P \cup C) = n(M) + n(P) + n(C)$   
 $- [n(M \cap P) + n(P \cap C) + n(C \cap M)] + n(M \cap P \cap C)$   
 $\Rightarrow 50 = 37 + 24 + 43$   
 $- [n(M \cap P) + n(M \cap C) + n(P \cap C)] + n(M \cap P \cap C)$   
 $\Rightarrow n(M \cap P) + n(M \cap C) + n(P \cap C)$   
 $= 54 + n(M \cap P \cap C)$

Given,

$$n(M \cap P) + n(M \cap C) + n(P \cap C) \leq 68 \text{ [For largest]}$$

$$54 + n(M \cap P \cap C) \leq 68$$

$$n(M \cap P \cap C) \leq 14$$

7. (b)  $\{1, 7\}$  This must be in subset so remaining will be  $\{2, 3, 4, 5, 6\}$   
 Now each element in subset will have two choice for being in subset or not.

So,

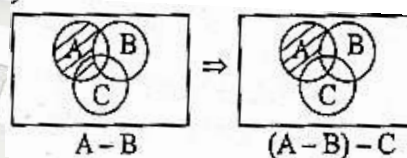
$$2^5 = 32$$

Of these excluding full set  $32 - 1 = 31$  (Proper subset)

8. (b) Option (a) and (c) are correct options it is clearly by De Morgan's law.

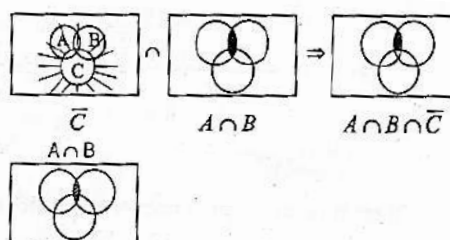
Now, option (b)

$(A - B) - C$  is only set  $A$  you can see it in given diagrams.



$(A - C) - (B - C)$  It is not possible. Hence, it is a wrong option.

Option (d)  $A \cap B \cap \bar{C}$



$$A \cap B \cap \bar{C} \subseteq A \cap B$$

Clearly, true

9. (b)  $n(A) = 200, n(B) = 300, n(A \cap B) = 100$   
 $n(\bar{A} \cap \bar{B}) = 300$   
 $n(\bar{A} \cap \bar{B}) = n(\overline{A \cup B}) = 300$   
 $n(A \cup B) = n(A) + n(B) - n(A \cap B)$   
 $n(A \cup B) = 200 + 300 - 100$   
 $n(A \cup B) = 400$   
 Given  $n(\overline{A \cup B}) = 300$   
 $n(U) - n(A \cup B) = 300$   
 $n(U) = 300 + n(A \cup B)$   
 $n(U) = 300 + 400 = 700 \Rightarrow n(x) = 700$

10. (c) Total number of readings =  $300 \times 5$

Also, total number of readings

= number of newspaper  $\times 60$

Let number of newspaper =  $n$

$$300 \times 5 = 60n$$

$$\Rightarrow n = 25$$



11. (d) Since,  $n(M \cup P) = 100, n(M) = 55, n(P) = 67$

$$n(M \cup P) = n(M) + n(P) - n(M \cap P)$$

$$100 = 55 + 67 - n(M \cap P)$$

$$n(M \cap P) = 22$$

$$\text{Passed in Physics only} = n(P) - n(M \cap P)$$

$$= 67 - 22 = 45$$

12. (d)  $n(A) = 150, n(B) = 180, n(C) = 210, n(D) = 240$

$$n(A \cap B) = n(B \cap C) = n(C \cap D) = n(A \cap D)$$

$$= n(A \cap C) = n(B \cap D) = 15$$

$$n(A \cap B \cap C) = n(B \cap C \cap D) = n(A \cap C \cap D)$$

$$= n(A \cap D \cap D) = 3$$

$$n(A \cap B \cap C \cap D) = \phi$$

$$n(A \cup B \cup C \cup D) = n(A) + n(B) + n(C) + n(D) - n(A \cap B)$$

$$- n(A \cap C) - n(A \cap D) - n(B \cap C) - n(B \cap D) - n(C \cap D) +$$

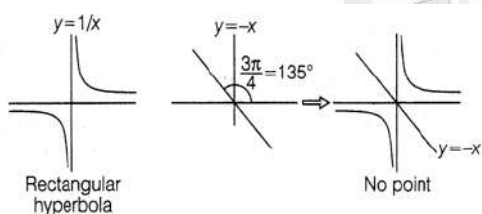
$$n(A \cap B \cap C) + n(A \cap B \cap D) + n(A \cap C \cap D) + n(B \cap$$

$$C \cap D)$$

$$- n(A \cap B \cap C \cap D)$$

$$= 150 + 180 + 210 + 240 - 6 \times 15 + 4 \times 3 - 0 = 702$$

13. (a)



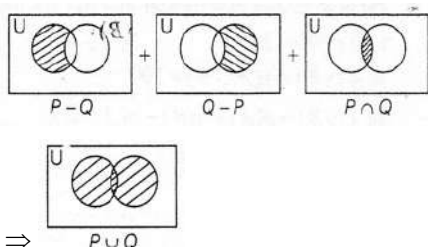
There is no any point of intersection, hence no solution.

14. (d) No materials of compiler.

$$= \text{Total books} - \text{Number of books with compiler.}$$

$$= 24 - 8 = 16.$$

15. (a)



16. (a)  $X = 4^n - 3n - 1$

$$n = 1 \Rightarrow 4 - 3 - 1 = 0$$

$$n = 2 \Rightarrow 16 - 6 - 1 = 9$$

$$X = \{0, 9, 54, \dots\}$$

Now,

$$Y = 9n - 9 = 9(n - 1)$$

$$n = 1 \Rightarrow 9(1 - 1) = 0$$

$$n = 2 \Rightarrow 9(2 - 1) = 9$$

$$n = 3 \Rightarrow 9(3 - 1) = 18$$

$$Y = \{0, 9, 18, \dots\}$$

$$X \subset Y$$

$$\text{Hence, } X \cup Y = Y$$

17. (Wrong) Question have two options right which is  $b$  and

(a)  $2 \in A \cup B$  implies that if  $2 \notin A$ , then  $2 \in B$  It is clearly right.

(b)  $\{2, 3\} \subseteq A$  implies that  $2 \subseteq A$  and  $\{2, 3\} \subseteq B$  Clearly, it is wrong.

(c)  $A \cup B \supseteq \{2, 3\}$  implies that  $\{2, 3\} \subseteq A$  and  $\{2, 3\} \subseteq B$  Clearly, it is wrong.

(d)  $\{2\} \in A$  and  $\{3\} \in A$  implies that  $\{2, 3\} \subseteq A$

$$\text{Ex. } A = \{1, 2, 3, 4\}, \{2\} \in A \text{ and } \{3\} \in A$$

Implies  $\{2, 3\} \subseteq A$

It is clearly right.

18. (b) According to question,  $2^m - 2^n = 56$ .

For  $m = 6, n = 3$  above satisfies. so  $m = 6, n = 3$ .

19. (b) First of all we choose set  $B$  from set  $S$  total number of ways to select set  $B = {}^nC_r$ .

Where  $0 \leq r \leq n, r$  is the number of element in set  $B$ .

Now, we select set  $A$  which is subset of set  $B$ .

Total number of ways to select set  $A = 2^r$

Now, total number of ways to select pair  $(A, B)$

$$\sum_{r=0}^n {}^nC_r (2^r) = \sum_{r=0}^n 2^r {}^nC_r$$

$$= ({}^nC_0 + 2 {}^nC_1 + 2^2 {}^nC_2 + \dots + 2^n {}^nC_n)$$

$$= 3^n \text{ [Using binomial series]}$$

20. (b) Number of elements of power set

$$= 2^{(\text{number of elements that set})}$$

$$\text{So, } n(S) = 2$$

$$\text{So, } n(P) = 2^2 = 4$$

21. (c)  $n(A) = 4, n(B) = 2, n(A \times B) = 8$

Number of subset with at least three elements = Total subsets - (number of subsets with one or two element)

$$= 2^8 - ({}^8C_1 + {}^8C_2)$$

$$= 256 - (8 + 28) = 256 - 36 = 220$$

Lastly we will also neglect  $\{\}$  (null set), so 219 subsets.



**22. (b)** Only exactly one subject

$$\begin{aligned} & [n(M) + n(L) + n(B)] - 2[n(M \cap L) + n(L \cap B) \\ & + n(B \cap M)] + 3n(M \cap L \cap B) \\ & = [32 + 38 + 30] - 2[7 + 10 + 8] + 3[5] \\ & = 100 - 2[25] + 15 = 100 - 50 + 15 = 65 \end{aligned}$$

**23. (d)**  $n(P_1) + n(P_2) + \dots + n(P_{30}) = 5 \times 30 = 150$

$$n(Q_1) + n(Q_2) + \dots + n(Q_n) = 3 \times n = 3n$$

Let set  $S$  has  $k$  element then,

$$k = \frac{150}{10} = 15 \text{ and } k = \frac{3n}{9} = \frac{n}{3} \Rightarrow 15 = \frac{n}{3} \Rightarrow n = 45$$

**24. (a)**

$$n(U) = 50, n(\text{Hitavad}) = 30, n(\text{Hindustan}) = 35,$$

$$n(\overline{\text{Hitavad} \cup \text{Hindustan}}) = 10$$

$$n(\text{Hitavad} \cup \text{Hindustan}) = 50 - 10 = 40$$

$$n(\text{Hitavad} \cup \text{Hindustan})$$

$$= n(\text{Hitavad}) + n(\text{Hindustan}) -$$

$$n(\text{Hitavad} \cap \text{Hindustan})$$

$$40 = 30 + 35 - n(\text{Hitavad} \cap \text{Hindustan})$$

$$n(\text{both}) = 65 - 40$$

$$n(\text{both}) = 25$$

**25. (a)** For  $x = 1$

$$A = \{0\}, B = \{0\}$$

For  $x = 2$

$$A = \{9\}, B = \{9\}$$

For  $x = 3$

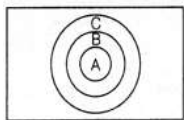
$$A = \{54\}, B = \{18\}$$

So,  $A \subset B$ .

**26. (b)**  $P(A) = \{\phi, \{x\}, \{y\}, \{z\}, \{x, y\}, \{y, z\}, \{z, x\}, \{x, y, z\}\}$

So, number of subsets in power set of  $A$  is  $2^3 = 8$

**27. (a)** Cardinality of  $C$



**28. (d)**  $A \Delta B = (A \cup B) - (A \cap B) = (A - B) \cup (B - A)$

$$A \cup B = \{1, 2, 3, 4, 5\}, A \cap B = \{3, 4\}$$

$$(A - B) = \{1, 2\}, (B - A) = \{5\}$$

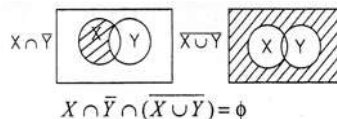
$$(A - B) \cup (B - A) = A \Delta B = \{1, 2, 5\}$$

$$n(A \cup B) = 5, n(A \cap B) = 2, n(A \Delta B) = 3$$

$$= 5 \times 2 \times 3 = 10 \times 3 = 30$$

**29. (b)** Refer to solution 21.

**30. (c)**



**31. (c)**  $n(A) = m; n(B) = n$

$$|P(A)| - |P(B)| = 112$$

$$\Rightarrow 2^m - 2^n = 112$$

$$\Rightarrow m = 7; n = 4$$

$$2m - n = 1$$

$$14 - 4 \neq 1$$

**32. (a)**

$$n(U) = 200$$

$$n(T) = \frac{60}{100} \times 200 = 120$$

$$n(C) = \frac{72}{100} \times 200 = 144$$

$$n(T \cap C) = x$$

$$(m \leq x \leq n) \quad n = \max, m = \min$$

$$n(T \cap C) \text{ is max when } T \subset C$$

In that case

$$n(T \cap C) = n(T) = n = 120$$

$$\text{Hence, } n = 120$$

$$n(T \cap C) \text{ is minimum when } n(T \cup C) = n(U)$$

$$n(T \cup C) = n(T) + n(C) - n(T \cap C)$$

$$200 = 120 + 144 - n(T \cap C)$$

$$n(T \cap C) = 64 = m$$

$$n - m = 120 - 64 = 56$$

**33. (a)** Let  $A$  and  $B$  be two sets such that

$$A \cap X = B \cap X = \phi \text{ and } A \cup X = B \cup X \text{ for some set } X.$$

$$\text{For } A = B$$

$$A = A \cap (A \cup X)$$

$$= A \cap (B \cup X)$$

$$\because [A \cup X = B \cup X]$$

$$= (A \cap B) \cup (A \cap X)$$

$$\because [\text{Distributive law}]$$

$$= (A \cap B) \cup \phi$$

$$\because [A \cap X = \phi]$$

$$= A \cap B \quad \dots \dots \dots (i)$$

$$\text{Now, } B = B \cap (B \cup X)$$

$$= B \cap (A \cup X)$$

$$[\because A \cup X = B \cup X]$$

$$(B \cap A) \cup (B \cap X)$$

$$[\text{Distributive law}]$$

$$= (B \cap A) \cup \phi$$

$$[\because B \cap X = \phi]$$

$$= B \cap A = A \cap B \quad \dots \dots \dots (ii)$$

Hence, from Eqs. (i) and (ii), we get

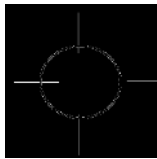
$$A = B$$

**34. (a)**  $X = \{(x, y): x^2 + 2y^2 = 3\} \times Y = \{(x, y): x > y, x, y \in \mathbb{Z}\}$



$$x^2 + 2y^2 = 3$$

$$\frac{x^2}{3} + \frac{y^2}{3/2} = 1$$



$$X = \{(1,1)(-1,1)(1,-1)(-1,-1)\}$$

$$Y = \{x(2,-x)(x,y)\}$$

$$X \cap Y = \{(1,-1)\}$$

option (a)

**35. (c)**  $x^2 - 2^y$

Solution of the equation

$$x^2 = 2^y \text{ are}$$

$$(1, 0) \text{ and } (2, 2)$$

The cardinality of the set C is 2 Ans.

**36. (d)**  $n(M \cup P \cup C) = 50$        $n(M) = 37, n(P) = 24$

$$n(C) = 43, n(M \cap P) = 19, n(M \cap C) = 29$$

$$n(P \cap C) = 20$$

$$n(M \cup P \cup C) = n(M) + n(P) + n(C) - n(M \cap P) - n(P \cap C)$$

$$- n(C \cap M) + n(P \cap C \cap M)$$

$$50 = 37 + 24 + 43 - 19 - 20 - 29 + n(P \cap C \cap M)$$

$$n(P \cap C \cap M) = 17$$

**37. (b)**  $C = \{x | \sin 4x = \frac{1}{2}, x \in (-9\pi, 3\pi)\}$

$$x \in (-9\pi, 3\pi)$$

$$4x \in (-36\pi, 12\pi)$$

$$\text{for } (0 \text{ to } 2\pi) \text{ two values of } x \text{ for } \sin 4x = \frac{1}{2}$$

So the interval of  $(-36\pi \text{ to } 12\pi)$  48 values of  $x$  are occurred.

Hence 48 values of  $x$ .

**38. (b)**

For  $n \in \mathbb{N}$ ,

$$5^n = (1 + 4)^n = 1 + 4n + \sum_{k=2}^n \binom{n}{k} 4^k,$$

and every term  $\binom{n}{k} 4^k$  ( $k \geq 2$ ) is divisible by 16 so

$$5^n - 4n - 1 \equiv 0 \pmod{16},$$

hence  $5^n - 4n - 1 \in B$  for all  $n$ . Thus  $A \subset B$ .

But  $32 \in B$  and no  $n$  gives  $5^n - 4n - 1 = 32$

(values: 0, 16, 112, ...), so  $B \not\subset A$ .

Therefore the correct choice is (b)  $A \subset B$ .

**39. (c)**

R: Countries with at least one river.

M: countries with at least one mountain.

D: Countries with at least one desert.

$D^c$ : Countries that do not have a desert.

countries that have either a mountain or a river = (RUM)

"But does not have a desert in it" =  $D^c = (RUM) \cap D^c$

Ans: option (C)  $(RUM) \cap D^c$